CSC3041F 2026
Concession Test

Automated Planning and Control

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Notes

1. Answer all questions
2. Test Time: 18h00-18h45, Tuesday, 24th of March, 2026
3. Test Duration: 45 minutes
4. There are six (6) questions, for a total of 35 marks
5. The **bold** type faced numbers in parentheses indicate the number of points for (sub)questions.

Read through the entire test paper before you start writing so that you can answer the easy questions first. The relative importance of the questions is indicated by the marks that are allocated to them.

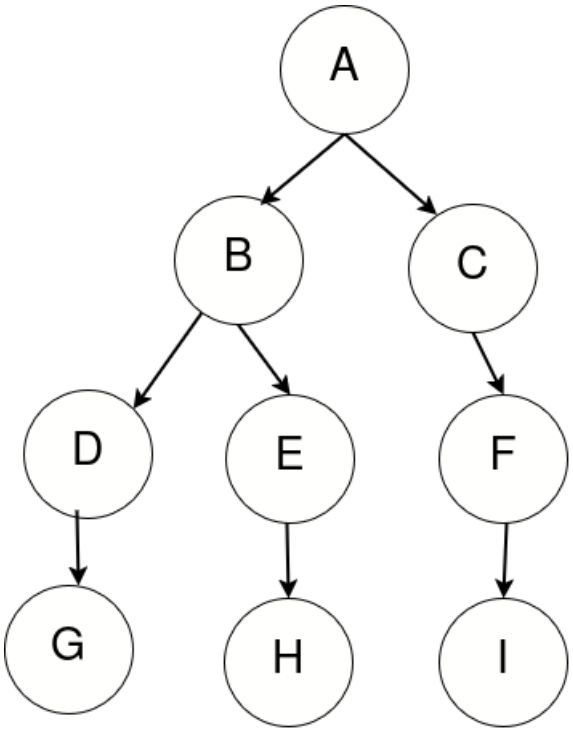


Figure 1: Visualisation of positions that an agent can occupy for question 2

Section 1 - Search

Question 1 [4 marks]

1. State whether the following statement is True/False: **[1]**

   In tree search, we check for and discard redundant paths by maintaining an explicit data structure for the history

2. Explain the difference between the Monte Carlo search and Monte Carlo tree search algorithms **[2]**

3. Explain why constructing a complete game tree is computationally infeasible for a chess-playing agent? **[1]**

Question 2 [6 marks]

Consider a scenario where there is a search agent starting at Location **A** with the objective of reaching Goal **I**. The available positions and paths between them are illustrated in Figure 1. Using the logic defined in Algorithm 1, answer the following questions:

Algorithm 1 Mystery Algorithm

```
1: function MysteryAlgo1(problem) returns a solution node or failure
2:     for depth = 0 to ∞ do
3:         result ← MysteryAlgo2(problem, depth)
4:         if result ≠ cutoff then return result
5:         end if
6:     end for
7: end function
8:
9: function MysteryAlgo2(problem, ℓ) returns a node or failure or cutoff
10:    frontier ← a LIFO queue (stack) with Node(problem.Initial) as an element
11:    result ← failure
12:    while not Is-Empty(frontier) do
13:        node ← Pop(frontier)
14:        if problem.Is-Goal(node.State) then return node
15:        end if
16:        if Depth(node) > ℓ then
17:            result ← cutoff
18:        else if not Is-Cycle(node) then
19:            for each child in Expand(problem, node) do
20:                add child to frontier
21:            end for
22:        end if
23:    end while
24:    return result
25: end function
```

1. Explain what Algorithm 1 does and how it works. **[1]**

2. Demonstrate that an agent that uses Algorithm 1 to find the goal (i.e., **I**) from **A** can never reach it if the depth value can never exceed 0. Show all intermediate steps and logic used to reach this conclusion. **[5]**

Section 2 - Reinforcement learning

Question 3 [5 marks]

For each question, choose the single best answer from the options provided.

1. Why is an ϵ-greedy policy used in reinforcement learning? **[1]**
   (a) To maximise immediate reward
   (b) To ensure convergence by sufficiently sampling different actions
   (c) To reduce variance
   (d) To avoid bootstrapping
   (e) To control the rate of learning

2. Why is TD learning often preferred over Monte Carlo? **[1]**
   (a) It does not require rewards
   (b) It can learn online from incomplete episodes
   (c) It avoids bootstrapping
   (d) It guarantees zero variance
   (e) It requires fewer states

3. Which statement correctly distinguishes between V^* and optimal policies? **[1]**
   (a) Both V^* and the optimal policy are unique.
   (b) Neither V^* nor the optimal policy is unique.
   (c) V^* may not be unique, but optimal policies are.
   (d) V^* is unique, but multiple optimal policies may exist.
   (e) Uniqueness depends on whether the MDP is deterministic.

4. What is the primary purpose of experience replay in DQN? **[1]**
   (a) To ensure updates use the most recent transitions for faster convergence
   (b) To approximate the on-policy distribution of the current policy
   (c) To decorrelate training data and reuse past experiences for more stable updates
   (d) To prioritise transitions with the largest temporal-difference error
   (e) To reduce the need for a target network

5. A Deep Q-Network is being trained. For the next state s', the network outputs:

   Behaviour network $Q(s^{'}, a_1) = 7$, $Q(s^{'}, a_2) = 5$

   Target network $Q_{target}(s^{'}, a_1) = 4$, $Q_{target}(s^{'}, a_2) = 6$

   The agent takes action a_1 in state s, receives reward $r = 3$, and transitions to s'. After one update step, what value is used as the target in the loss for updating $Q(s, a_1)$ if we set $\gamma = 0.5$? **[1]**

(a) 6.5
(b) 6
(c) 5
(d) 5.5
(e) 4.5

Question 4 [6 marks]

State whether each of the following statements are true or false. Give a reason for your answer.

1. Temporal Difference learning updates all state values at every timestep. [**2**]
2. Value Iteration can be seen as performing policy improvement at every update. [**2**]
3. Acting greedily with respect to a value function yields an optimal policy. [**2**]

Question 5 [6 marks]

For each scenario, state whether on-policy or off-policy RL is more appropriate, and briefly justify your choice.

1. Self-driving car in a simulator [**2**]
2. Recommender system using only collected data. [**2**]
3. Robot learning tasks in a virtual environment. [**2**]

Question 6 [8 marks]

Consider the code in Figure 2 and answer the following questions:

```
for iteration in range(K):
    for s in states:
        a = pi[s]
        V[s] = sum(
            P(s_prime, s, a) * (R + gamma * V[s_prime])
            for s_prime in states
        )
```

Figure 2:

1. Is the code synchronous or asynchronous? Explain your answer. **[1]**

2. For the policy iteration algorithm, which Bellman equation is used during the Policy Evaluation phase, and why? **[2]**

3. For the policy iteration algorithm, which Bellman equation is used during the Policy Improvement phase, and why? **[2]**

4. Given the following information:

   States: s_1, s_2

   Discount $\gamma = 0.5$

   $V^{\pi}(s_1) = 5.0, V^{\pi}(s_2) = 3.0$ (from the previous step)

s	a	s'	$P(s' \mid s, a)$	$R(s, a, s')$
s_1	a_1	s_1	0.5	2
s_1	a_1	s_2	0.5	4
s_1	a_2	s_1	0.75	3
s_1	a_2	s_2	0.25	1

   Compute the Q-values for s_1 and determine the new policy action. **[3]**